E-Commerce Orders Analysis

Final Project Report | Jan – Jun 2025 | Python · SQL Server · Power BI

1. Overview

This report answers the seven business questions using the cleaned order data (321 order records, 6 cities, 4 categories, 6 products). Data was cleaned in Python, queried in SQL Server and visualised in Power BI. Sales means Net Amount (after discount); profit is 20% of Net Amount.

Total Orders	Total Sales	Total Profit	Avg. Order Value
321	₹32,55,623	₹6,51,124	₹10,142

2. Answers to the Business Questions

Q1. Which products generate the highest revenue?

Product	Sales (₹)
Laptop	6,53,570
Chair	6,18,839
Mouse	5,98,230
Bottle	5,39,841
Watch	4,69,805
Shoes	3,75,336

Laptop is the top revenue product, closely followed by Chair. Shoes earns the least.

Q2. Which categories are the most profitable?

Category	Profit (₹)
Electronics	2,50,360
Fashion	1,69,028
Furniture	1,23,767
Home	1,07,968

Electronics is the most profitable category (about 38.45% of total profit).

Q3. Which payment methods are preferred?

Payment Mode	Orders	Sales (₹)
COD	77	6,87,466
Wallet	75	7,39,710
Net-Banking	68	6,89,693
UPI	56	6,14,467
Card	45	5,24,284

COD has the most orders, with Wallet and Net-Banking very close.

Wallet brings the highest sales value and Card is the least used.

Q4. How many orders were cancelled?

81 orders (about 25% of all orders).

Other statuses: Delivered 83, Returned 83, Pending 74. Only **26% of orders are delivered**, so cancellations and returns together (50%) are a major concern.

Q5. Which cities contribute the most to sales?

City	Sales (₹)
Bengaluru	7,66,916
Pune	7,13,440
Delhi	5,09,770
Lucknow	4,63,146
Mumbai	4,26,893
Jaipur	3,75,455

Pune and Bengaluru together contribute about 46% of total sales.

Q6. Which months perform the best?

Month	Sales (₹)
May	8,02,293
June	6,58,385
February	5,90,827
April	5,52,956
January	3,66,658
March	2,84,501

May is the best month and March the weakest.

Q7. What is the average order value?

₹10,142 (Total Net-Amount ÷ 321 orders with amounts).

3. Key Insights & Recommendations

- Focus marketing and stock on Laptops and Electronics – top in both revenue and profit.
- Investigate the high cancellation (25%) and return (25%) rates; improve delivery, product quality and COD confirmation.
- Expand in Pune and Bengaluru; run campaigns in Jaipur and Mumbai to lift weaker markets.
- Plan promotions and inventory around the May–June peak; run offers to lift March.
- Encourage UPI/Card with cashback, since COD orders carry higher cancellation risk.